Classifying Mental Health-Related Text with Natural Language Processing

Sabriya F. Sowers

Northwest Missouri State University, Maryville MO 64468, USA
S575583@nwmissouri.edu and sabriyasowers@gmail.com

Abstract. This capstone project investigated the application of Natural Language Processing (NLP) and supervised machine learning techniques for the classification of mental health-related social media text. A publicly available dataset obtained from Kaggle containing posts labeled as Neutral, Depression, and Suicidal Tendencies was used to develop predictive classification models. The data underwent validation, preprocessing, exploratory data analysis, and TF-IDF feature engineering before four supervised machine learning algorithms, Logistic Regression, Linear Support Vector Machine (Linear SVM), Random Forest, and Multinomial Naïve Bayes, were trained and evaluated. Model performance was assessed using accuracy, weighted precision, weighted recall, weighted F1-score, and confusion matrix analysis. Logistic Regression and Linear Support Vector Machine achieved the highest overall classification accuracy (88.40%), while Logistic Regression achieved the highest weighted precision and provided interpretable probability estimates. Based on these evaluation metrics, Logistic Regression was selected as the final predictive model. The confusion matrix further demonstrated that most posts were correctly classified, although distinguishing some Suicidal posts from Neutral posts remained the most challenging classification task. Overall, the results demonstrate that combining text preprocessing, TF-IDF feature engineering, and supervised machine learning provides an effective and reproducible approach for mental health text classification while establishing a foundation for future research involving the automated analysis of mental health-related text data.

Keywords: Natural Language Processing (NLP) · Text Classification · Machine Learning · Mental Health · Suicide Risk Detection

1 Introduction

For this capstone project, I focused on the field of Natural Language Processing (NLP). I selected this domain because my graduate coursework in Web Mining and Applied Natural Language Processing provided a strong foundation in NLP techniques, and I wanted the opportunity to apply those skills to a real-world dataset. I have also always considered depression and suicide to be important areas of mental health research, and I believe data analytics has the potential to support researchers by identifying meaningful language patterns that may contribute to earlier awareness and intervention efforts.

To identify an appropriate dataset, I used Google Dataset Search to evaluate several publicly available options [2]. I selected the Suicide Sentiment Analysis Dataset from Kaggle, which was originally developed for the published research study Enhanced Feature Engineering Approach for Suicidal Risk Identification (EFASRI) by Hajam et al. [3]. The dataset was downloaded from Kaggle [4]. The dataset contains text labeled as Neutral, Depression, and Suicidal Tendencies, making it well-suited for developing and evaluating NLP classification models. This project applied NLP techniques and machine learning algorithms to develop and evaluate models capable of classifying text into these three categories.

The primary objective of this project is to determine whether machine learning models can accurately classify text into the categories of Neutral, Depression, and Suicidal Tendencies. I find this problem particularly interesting because it combines NLP with a meaningful real-world application, allowing machine learning techniques to analyze human language in ways that may support mental health research. This problem is significant because social media have become a major form of communication and self-expression, and online text may provide early indicators of an individual's emotional state through the language they use. Although machine learning should never replace the evaluation of a qualified mental health professional, text classification models have the potential to assist researchers and organizations by identifying language patterns that may warrant further attention. This project demonstrates how data analytics can be applied to support mental health research and contribute to the development of tools that can aid early intervention efforts.

2 Data

2.1 Dataset Source and Description

The dataset used for this project is the Suicide Sentiment Analysis Dataset, which was obtained from Kaggle. The dataset was originally developed for the research study *Enhanced Feature Engineering Approach for Suicidal Risk Identification (EFASRI)* by Hajam et al. and contains social media text classified into three categories: Neutral, Depression, and Suicidal Tendencies.

The dataset was provided as three CSV files representing the Neutral, Depression, and Suicidal Tendencies classes. Because the dataset was downloaded directly from Kaggle, no web scraping or additional data extraction procedures were required prior to analysis. The three datasets were then imported into Python using Pandas for initial exploration, validation, and preparation.

The original dataset consisted of two primary attributes: TEXT and Label. During preprocessing, additional attributes were created to preserve intermediate processing stages, including `TEXT_ORIGINAL`, `TEXT_REPAIRED`, `TEXT_NORMALIZED`, and `TEXT_PROCESSED`. The `TEXT_PROCESSED` attribute serves as the primary input for feature engineering and machine learning model development, while the

Label attribute represents the target classification category of Neutral (0), Depression (1), or Suicidal Tendencies (2).

2.2 Dataset Summary

Following data validation and preprocessing, the final dataset used for analysis contained 15,774 social media posts representing three target classes: Neutral, Depression, and Suicidal Tendencies. The dataset includes the original class labels, intermediate preprocessing attributes, and the final `TEXT_PROCESSED` attribute, which serves as the primary input for feature engineering and machine learning model development. The following tables and figures summarize the characteristics of the final dataset and present the exploratory data analysis performed prior to model development.

3 Methodology

3.1 Data Validation and Preparation

Each dataset was inspected to validate its structure by reviewing the column names, data types, record counts, missing values, and duplicate records. During validation, the Depression dataset contained an unnecessary column (`Unnamed: 2`) that resulted from the original file export. Because the column contained no meaningful analytical information, it was removed prior to merging the datasets.

The original CSV files also contained character encoding inconsistencies. To correctly import the text into Python, the datasets were loaded using the `latin1` encoding. A small number of records containing character encoding artifacts were identified during validation. Because most affected records remained readable, these records were retained and will be evaluated further during the text preprocessing phase.

Missing values were then evaluated. The Neutral dataset contained missing values in the **Label** field even though every record represented the Neutral class. These missing labels were assigned the value 0 to maintain consistency with the dataset's intended classification. Records containing missing values in the **TEXT** field were removed because machine learning classification requires textual input for model training and prediction.

After validation and cleaning, the three datasets were merged into a single dataset using the Pandas Python library and exported as a cleaned CSV file. The final cleaned dataset contained 17,699 records and two attributes: TEXT and Label. The Label attribute serves as the dependent variable representing the target mental health classification, while the TEXT attribute serves as the independent variable containing the social media posts used for Natural Language Processing and machine learning classification

Fig. 1 summarizes the overall data curation workflow used to prepare the dataset for exploratory data analysis and machine learning model development.

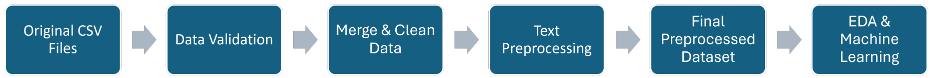

Fig. 1. Data curation workflow illustrating the validation, cleaning, preprocessing, and preparation of the dataset for exploratory data analysis and machine learning model development.

3.2 Data Preprocessing

Following data validation and preparation, the text data underwent several preprocessing steps to improve consistency and prepare the dataset for Natural Language Processing (NLP) and machine learning. During text quality assessment, four unique text values (".", "Alone.", "I hate myself.", and "I'm tired") were identified with conflicting class labels. Because identical text assigned to different target classes introduces ambiguity during supervised machine learning, these conflicting records were removed prior to further preprocessing. Duplicate posts with matching labels were also removed to prevent redundant observations from biasing the classification models.

Character encoding artifacts present in the original source dataset were then inspected and repaired using the `ftfy` library. While many encoding issues were corrected automatically, some artifacts resulted from permanent corruption in the original source data and could not be fully recovered. Because the surrounding text remained readable and meaningful, these records were retained rather than removed.

After character encoding repair, the text was normalized to improve consistency across all observations. Regular expressions were used to remove HTML tags, URLs, email addresses, user mentions, and common social media editorial markers such as `EDIT`, `UPDATE`, `TL;DR`, `ETA`, and `RT`. Additional normalization included converting text to lowercase, replacing numeric values with a common `number` token, removing punctuation, and standardizing whitespace.

The normalized text was then processed using the spaCy Python natural language processing library. Tokenization and lemmatization were applied to reduce words to their base forms, while common stop words were removed to reduce noise in the dataset. However, important negation terms such as *no*, *not*, and *never* were intentionally preserved because they can substantially change the meaning and sentiment of a statement. For example, the phrases "I am happy" and "I am not happy" differ by only one word but express opposite meanings.

Following preprocessing, documents containing no meaningful processed text were identified and removed because they did not provide useful textual features for classification. The resulting preprocessed dataset contained 15,774 records and was exported for use during exploratory data analysis, feature engineering, and machine learning model development.

No additional data extraction or transformation was required beyond downloading the publicly available dataset. The dataset was imported directly into the analysis environment for preprocessing and feature engineering.

4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is the process of examining and summarizing a dataset to better understand its structure, identify patterns, detect anomalies, and inform subsequent analysis [8]. EDA is a critical step in the data science workflow because it provides insight into the characteristics of a dataset before feature engineering and machine learning model development. By exploring the data prior to model training, potential issues can be identified, data quality can be verified, and meaningful patterns can be discovered that guide subsequent analytical decisions.

Several exploratory data analysis techniques were applied to the preprocessed mental health text dataset, including dataset validation, class distribution analysis, descriptive statistics, word count analysis, word frequency analysis, bigram analysis, and qualitative visualization through a word cloud. These techniques were selected to evaluate the quality of the cleaned dataset, understand the distribution of the target classes, examine differences in text characteristics, and identify common language patterns that may contribute to text classification.

The exploratory analysis began by validating the final preprocessed dataset through examination of the dataset dimensions, data types, missing values, duplicate records, and class distribution. The resulting dataset contained 15,774 social media posts classified into three categories: Neutral, Depression, and Suicidal Tendencies. Validation confirmed that the preprocessing pipeline successfully produced a complete dataset without missing values or duplicate records, indicating that the data were suitable for machine learning.

Class distribution analysis showed that the three classes represented approximately 27% to 38% of the dataset, suggesting that class imbalance was not severe enough to require oversampling or undersampling techniques. Descriptive statistics and boxplot analysis of text length revealed clear differences among the classes. Depression posts contained the longest messages, with a median of 114 words, followed by Suicidal Tendencies with a median of 61 words, while Neutral posts contained a median of only 9 words. Greater variation in post length was also observed within the Depression and Suicidal Tendencies classes, indicating that individuals discussing mental health concerns generally produced longer and more detailed narratives.

Word frequency analysis confirmed that preprocessing successfully removed common stop words while preserving important negation terms such as "not" and "no," which retain meaningful context in mental health text. Frequently occurring words including "know," "want," "life," and "feel" reflected the conversational and emotionally expressive nature of the dataset. Bigram analysis further demonstrated that contextual information had been preserved during preprocessing by identifying common phrases such as "feel like," "want die," "commit suicide," and "not understand." These multi-word expressions provide greater contextual meaning than individual words and may improve subsequent feature engineering and machine learning performance.

A qualitative word cloud was also generated for the Depression class to visualize commonly occurring vocabulary. Although the visualization provided an effective summary of the dominant themes within Depression posts, the word frequency tables and bigram analysis yielded more meaningful analytical insights regarding language usage and contextual patterns.

The exploratory analysis demonstrated that the preprocessed dataset was well suited for text classification. Validation confirmed that the dataset was complete and free of missing values and duplicate records, while the relatively balanced class distribution indicated that severe class imbalance was not expected to affect model training. The analysis also revealed meaningful differences in text length across the three classes, with Depression and Suicidal Tendencies posts generally containing longer and more detailed narratives than Neutral posts. Word frequency and bigram analyses further showed that the preprocessing pipeline successfully preserved important contextual language patterns that are expected to contribute valuable information during feature engineering.

In addition to these findings, this phase demonstrated that not every exploratory technique contributes equally to understanding the data. Several exploratory visualizations were evaluated during development but ultimately omitted because they did not provide additional insight beyond the retained analyses. This process emphasized that effective exploratory data analysis involves selecting techniques that provide meaningful information rather than maximizing the number of visualizations. The findings from this phase establish a strong foundation for the subsequent feature engineering process, where the textual data will be transformed into numerical representations suitable for machine learning algorithms.

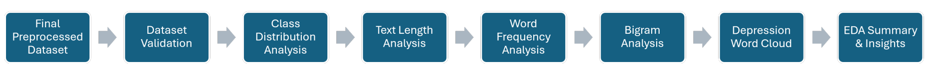

Fig. 2. Exploratory data analysis workflow for the preprocessed mental health text dataset.

4.1 Feature Engineering

Following text preprocessing, the cleaned text was transformed into numerical features suitable for supervised machine learning. Term Frequency–Inverse Document Frequency (TF-IDF) vectorization was selected because it represents text data by assigning higher weights to words that are more informative within individual documents while reducing the influence of commonly occurring terms across the corpus.

The dataset was divided into training (80%) and testing (20%) subsets using stratified sampling to preserve the class distribution. The TF-IDF vectorizer was fit using only the training data and then applied to both the training and testing datasets. This approach ensured that the testing data remained independent during model development and provided a more realistic evaluation of model performance. The resulting sparse feature matrices served as the input for the predictive modeling phase.

4.2 Model Development

Four supervised machine learning algorithms were selected to provide a representative comparison of commonly used approaches for text classification: Logistic Regression, Linear Support Vector Machine (Linear SVM), Random Forest, and Multinomial Naïve Bayes. Logistic Regression was selected because it is a widely used baseline classifier for natural language processing and provides interpretable probability estimates. Linear SVM was included because of its strong performance on high-dimensional sparse text data generated through TF-IDF feature extraction. Random Forest was selected to evaluate whether an ensemble learning approach could improve classification performance by capturing more complex decision boundaries. Multinomial Naïve Bayes was included because it is a traditional probabilistic algorithm that has historically performed well on document classification tasks and serves as an important benchmark for comparison.

Following feature engineering, the processed text was transformed into TF-IDF feature vectors using the previously saved vectorizer. The transformed training data were then used to train each of the four machine learning algorithms. A fixed random seed (`random_state = 42`) ensured that the data partitioning remained reproducible throughout model development.

After training, each model generated predictions for the testing dataset. These predictions were subsequently used to evaluate and compare model performance using multiple classification metrics.

The complete implementation, including the data preprocessing, feature engineering, predictive modeling, and evaluation notebooks, is available in the accompanying GitHub repository [7].

4.3 Model Evaluation

The performance of each classification model was evaluated using multiple performance metrics, including accuracy, weighted precision, weighted recall, weighted F1-score, and detailed classification reports. These metrics provided a comprehensive assessment of overall model performance as well as class-level performance.

The evaluation results were compiled into a comparison table to identify the strongest-performing model. Logistic Regression and Linear SVM achieved the highest overall classification accuracy (88.40%). Although Linear SVM achieved a slightly higher weighted F1-score, Logistic Regression demonstrated the highest weighted precision while providing probability estimates and straightforward model interpretation. Based on the overall evaluation metrics, interpretability, and probability estimates, Logistic Regression was selected as the final predictive model and saved for future inference.

5 Results

The predictive modeling results were interpreted using two primary visualizations: a model performance comparison chart and a confusion matrix for the selected Logistic Regression model. The model performance comparison chart was selected because it provides a clear visual comparison of the overall accuracy, weighted precision, weighted recall, and weighted F1-score across all four supervised machine learning models, making it easier to identify the highest-performing model. The confusion matrix was selected because it provides a detailed view of the classification performance of the final model by illustrating both correct classifications and misclassifications for each mental health category. Together, these visualizations summarize overall model performance while also identifying specific strengths and areas for improvement in the final classifier.

Table 1 summarizes the evaluation metrics for the four supervised machine learning models and provides the quantitative evidence supporting model selection.

Table 1. Performance Comparison of Machine Learning Models

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Linear SVM	0.8840	0.8843	0.8840	0.8838
Logistic Regression	0.8840	0.8858	0.8840	0.8834
Random Forest	0.8621	0.8636	0.8621	0.8620
Multinomial Naïve Bayes	0.8162	0.8503	0.8162	0.8123

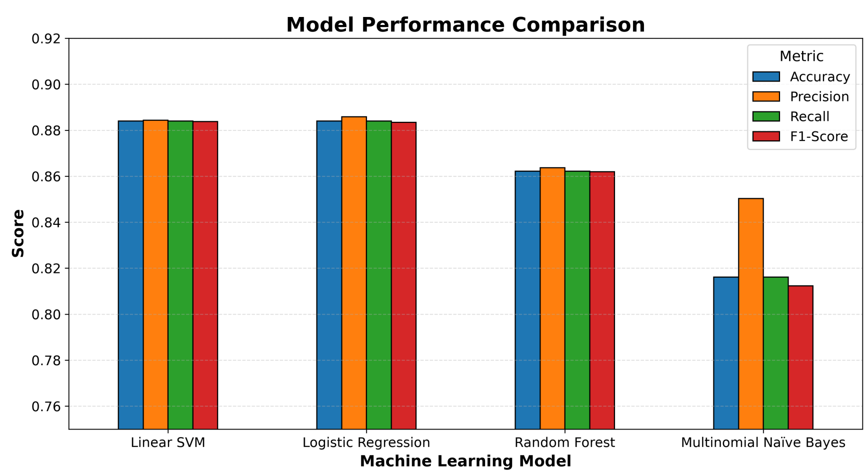

Fig. 3. Performance comparison of the four supervised machine learning models using accuracy, weighted precision, weighted recall, and weighted F1-score.

Figure 3 shows that Logistic Regression and Linear Support Vector Machine achieved the highest overall classification accuracy (88.40%), while Random Forest (86.21%) and Multinomial Naïve Bayes (81.62%) produced lower overall performance. Although Linear SVM achieved a slightly higher weighted F1-score, Logistic Regression achieved the highest weighted precision while matching the highest overall accuracy. These evaluation metrics indicate that Logistic Regression provided the best overall balance of predictive performance and interpretability for this dataset. Based on this evidence, Logistic Regression was selected as the final predictive model.

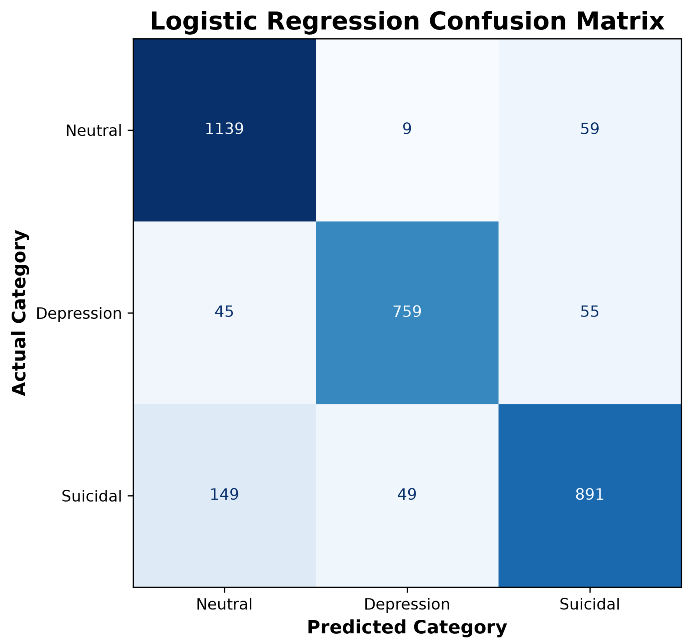

Fig. 4. Confusion matrix for the selected Logistic Regression model.

Figure 4 shows that the Logistic Regression model correctly classified most observations across all three mental health categories. The largest values appear along the main diagonal of the confusion matrix, indicating that most posts were classified correctly. This is important because it indicates that the model accurately distinguished among the three mental health categories for the majority of testing observations. Neutral and Depression posts achieved particularly strong classification performance, while the largest number of misclassifications occurred when Suicidal posts were predicted as Neutral (149 cases). This indicates that distinguishing some Suicidal posts from Neutral posts was the most challenging classification task for the model.

5.1 Data Inference

The evaluation metrics indicate that the TF-IDF features provided sufficient information for the machine learning models to distinguish among the three mental health categories. This inference is supported by the overall classification accuracies ranging from 81.62% to 88.40% across the four evaluated models. The strong performance of both Logistic Regression and Linear Support Vector Machine demonstrates that linear classification algorithms were effective for this text classification task. The confusion matrix further indicates that distinguishing some Suicidal posts from Neutral posts remained the most difficult aspect of the classification problem.

5.2 Statistical Conclusions

The statistical evidence demonstrates that Logistic Regression provided strong and balanced classification performance across the evaluation metrics while achieving an overall classification accuracy of 88.40%. Although Linear Support Vector Machine achieved a slightly higher weighted F1-score, Logistic Regression also achieved the highest weighted precision and matched the highest overall accuracy. Based on these evaluation metrics and its interpretability, Logistic Regression was selected as the final predictive model because it provided the best overall balance between predictive performance and interpretability for this dataset.

5.3 General Observations

Overall, the evaluation metrics and confusion matrix demonstrate that combining text preprocessing, TF-IDF feature engineering, and supervised machine learning provided an effective and reproducible approach for classifying the mental health text dataset used in this study. Because the majority of posts were correctly classified and Logistic Regression achieved the strongest overall balance of evaluation metrics, the predictive modeling workflow successfully identified meaningful language patterns within the dataset. The confusion matrix also identified the primary opportunity for improvement, as some Suicidal posts were misclassified as Neutral.

6 Limitations

Several limitations should be considered when interpreting the results of this study. First, the project used a publicly available dataset obtained from Kaggle that was originally developed from Twitter and Reddit posts for research purposes. Consequently, the language patterns represented in the dataset may not generalize to other social media platforms, communication channels, or populations.

Second, the machine learning models were trained to classify text into only three categories: Neutral, Depression, and Suicidal Tendencies. Real-world mental health discussions are often more nuanced and may not fit neatly into these predefined classes.

Finally, this project evaluated traditional supervised machine learning algorithms using TF-IDF feature engineering. While these approaches produced strong classification performance, more advanced deep learning models, such as transformer-based architectures, may achieve improved performance on complex language classification tasks.

7 Future Work

Although the selected Logistic Regression model demonstrated strong classification performance, several opportunities exist to extend this research. Future studies could evaluate transformer-based language models, such as BERT and RoBERTa, because these models capture contextual relationships between words rather than relying solely on word frequency, which may improve the classification of posts containing subtle language patterns, ambiguous wording, or other context-dependent expressions [1, 5].

Future research could also evaluate additional datasets collected from multiple social media platforms to determine whether the predictive models generalize across different populations, writing styles, and communication patterns. Using a more diverse collection of text data would provide greater confidence that the models perform consistently beyond the dataset used in this study.

Additional opportunities include investigating advanced hyperparameter optimization techniques to determine whether model performance can be further improved through systematic parameter tuning. Explainable artificial intelligence (XAI) methods could also be incorporated to improve model transparency by identifying which words and phrases contribute most strongly to each prediction, providing greater insight into the model's decision-making process [6]. Finally, expanding the classification task to include additional mental health conditions or evaluating longitudinal datasets could further increase the practical applicability of automated mental health text classification.

8 Conclusion

This study demonstrated the application of natural language processing and supervised machine learning techniques for the classification of mental health-related social media text. Following data validation, preprocessing, exploratory data analysis, TF-IDF feature engineering, and predictive modeling, four supervised machine learning algorithms were evaluated using standard classification metrics to identify the most effective predictive model.

The evaluation metrics and confusion matrix demonstrated that Logistic Regression and Linear Support Vector Machine achieved the strongest overall performance, with Logistic Regression selected as the final predictive model because it achieved the highest weighted precision while matching the highest overall classification accuracy and providing strong interpretability. The results also demonstrated that the selected model successfully distinguished among the three mental health categories while identifying the classification of some Suicidal posts as Neutral as the primary remaining challenge.

Overall, this project demonstrates that combining text preprocessing, TF-IDF feature engineering, and supervised machine learning provides an effective and reproducible approach for mental health text classification. The findings support the potential for predictive analytics to assist future research involving automated analysis of mental health-related text data while also providing a reproducible workflow that can serve as a foundation for future machine learning studies.

References

1. Devlin, J., Chang, M.W., Lee, K., Toutanova, K.: Bert: Pre-training of deep bidirectional transformers for language understanding. Proceedings of NAACL-HLT (2019)
2. Google: Google Dataset Search. https://datasetsearch.research.google.com/ (2026), accessed August 2026
3. Hajam, U.A., Rabani, S.T., Khanday, A.M.U.D., Khan, Q.R., Imran, A.S., Kastrati, Z.: Enhanced feature engineering approach for suicidal risk identification. Egyptian Informatics Journal (2023). https://doi.org/10.1016/j.eij.2023.04.003
4. Kaggle: Suicide Sentiment Analysis Dataset. https://www.kaggle.com/datasets/umar1103/suicide-sentiment-analysis-dataset (2026), accessed August 2026
5. Liu, Y., et al.: Roberta: A robustly optimized bert pretraining approach. arXiv preprint arXiv:1907.11692 (2019)
6. Lundberg, S.M., Lee, S.I.: A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems **30** (2017)
7. Sowers, S.F.: Mental Health Text Classification Using Natural Language Processing. https://github.com/ssowers2/capstone-nlp-mental-health (2026), gitHub repository. Accessed August 2026
8. Wickham, H., Grolemund, G.: R for Data Science: Import, Tidy, Transform, Visualize, and Model Data. O'Reilly Media, Sebastopol, CA (2017)